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HW0- Problem 1

a.) The algorithm is a prime checker that, given a number N, iterates over every number between(and including) 2 and N, checking if it is prime. This is done through verifying that a number has more than 2 divisors(normally 1 and itself for prime cases), where a modulo checking if the number, when divided, has a nonzero remainder. If that is the case, it is a divisor. Each prime number is output.

b.) pseudo code:

```
Input N
For each number(i) between(and including) 2 to N:
        For each number(j) between and including 1 to i:
                Divisorcount = 0
                Check if j is a divisor of i
                Divisorcount++
                If more than 2 divisors, terminate inner loop
        If more than 2 divisors, print out number as it’s prime
```

c.) Primes fundamentally mean a number has a divisor that isn't 1 or itself. Iterating over every possible number between 2 and our target, and checking if there’s 3 or more divisors will verify this.

d.) O(n^2)

e.) We check n values in both the inner and outer loops. Thus, as we scale up to bigger numbers, it should be n^2.